## Group composition and legend of task distribution:

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- Yuhsaku NAKAI
- Jayden TAN
- The group

## Provisional planning

Milestone 1: Setting up the common tools (Nov 6-12):

- Choice of programming language (about 1 day)
- Input data (instance) generator (about 4 days)
- USCP benchmark parser: unweighted instances E, CLR, CYC, with the known optimum or best known solution recorded (see the GitHub repository) (about 4 days)
- Setting up the common interface for all algorithms (about 4 days):
  - Input and output format
  - Timing
  - Solution verification

Milestone 2: First algorithms and validation (Nov 13-26):

- Brute force (about 3 days)
- Greedy (about 3 days)
- Dynamic programming (about 7 days)
- Branch & bound (about 10 days)
- Cross-checking tests: the exact algorithms (brute force, branch & bound, dynamic programming) must find the same optimal solution size (about 2 days)
- First experiments on generated instances (about 2 days)

Milestone 3: Advanced methods (Nov 27 - Dec 6):

- Randomized greedy (about 4 days)
- Local search (about 4 days)
- Simulated annealing (about 4 days)
- Fixed experimental protocol: same machines, same random seeds, repetitions (about 1 day)
- First evaluations on the benchmarks (about 3 days)

Milestone 4: Full experiments (Dec 7-10):

- Run and test all algorithms on the generated instances (about 2 days)
- Run and test all algorithms on at least 4 USCP benchmarks, then on a large part of the collection (about 3 days)
- Charts with error bars and analysis by instance type (about 3 days)

Milestone 5: Report and finalization (Dec 11-15):

- Report writing:
    - sections on their own algorithms (each member covers their own code) (about 2 days)
    - Report writing: introduction, protocol, results, analysis (about 4 days)
    - Report writing: actual planning, checklist, workload percentages (about 3 days)

- Code documentation, comments, installation README (each member covers their own code) (about 2 days)
- Cross-proofreading of the code and report clarity (about 1 day)
- Archive check and reproducibility test on a clean machine (about 2 days)

Milestone 6: Defense preparation (Dec 15-17)

- Preparing the slides and the presentation outline (about 1 day)
- Defense preparation (about 1 day, including rehearsal)

## List of the 7 algorithms:

- Brute force
- Branch & bound
- Greedy
- Dynamic programming
- Additional algorithm 1: Randomized greedy
- Additional algorithm 2: Local search
- Additional algorithm 3: Simulated annealing